

**William Stallings
Computer Organization
and Architecture
7th Edition**

**Chapter 2
Internal Memory**

What is Memory ?

- Physical devices used to store **programs** (sequences of instructions) or **data** on a **temporary** or **permanent** basis for use in a **computer**.

Two types of Memory

- **Secondary Memory**
- **Primary or Main Memory**

Secondary Memory (Hard Disk etc)

- **Non-volatile** and **persistent** in nature and is not directly accessed by a processor
- Secondary memory is accessed only by the primary or main memory
- **Slower** than P.M but **larger** in size and **cheap**
- Data can be stored for later retrieval
 - HDD, Tape Drives etc

Primary Memory (RAM etc)

- Computer memory that a processor or computer accesses **first or directly**
 - Allows a processor to access running execution applications and services
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- **Random Access Memory (RAM)**
 - **Read Only Memory (ROM)**

Semiconductor Memory Types

Memory Type	Category	Erasure	Write Mechanism	Volatility
Random-access memory (RAM)	Read-write memory	Electrically, byte-level	Electrically	Volatile
Read-only memory (ROM)	Read-only memory	Not possible	Masks	Nonvolatile
Programmable ROM (PROM)				
Erasable PROM (EPROM)	Read-mostly memory	UV light, chip-level	Electrically	
Electrically Erasable PROM (EEPROM)		Electrically, byte-level		
Flash memory		Electrically, block-level		

Semiconductor Memory

1. Read Write Memory(RWM)(RAM)
2. ROM (ROM, PROM)
3. RMM (EPROM,EEPROM)

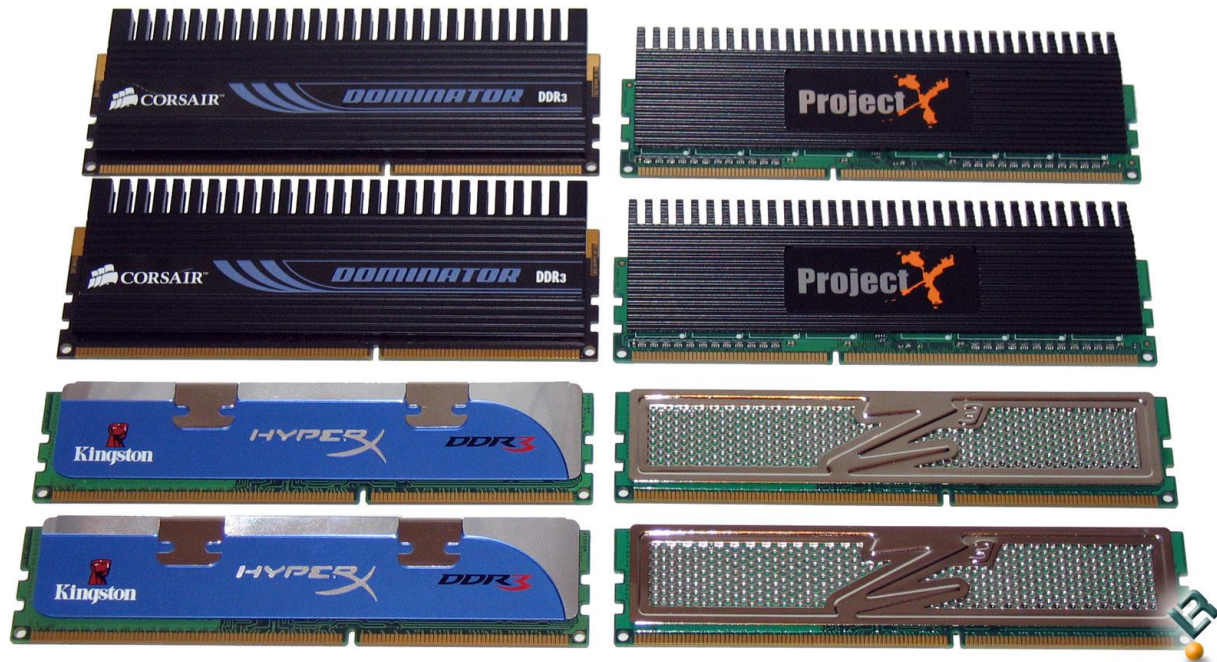
1.RAM

- ✓ Misnamed as all semiconductor memory is random access
- ✓ Read/Write
- ✓ Data is written electrically , electrically erasable
- ✓ Volatile
- ✓ Temporary storage

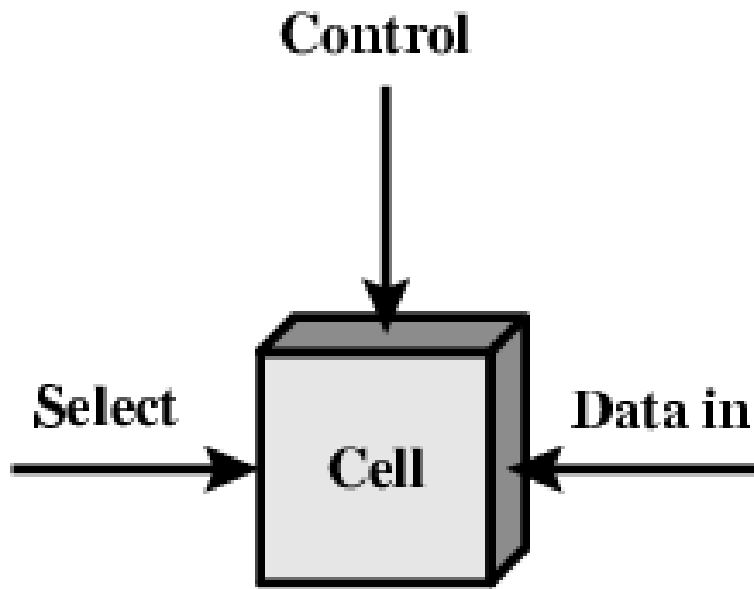
TWO TYPES OF RAM

- Static RAM
- Dynamic RAM

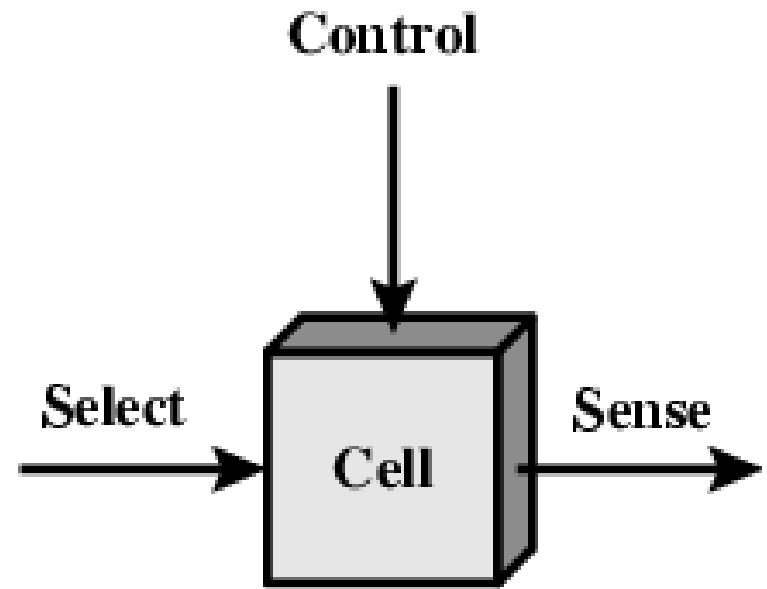
RAM: Random Access Memory



Memory Cell Operation



(a) Write



(b) Read

Types of RAM

1.DYNAMIC RAM:

- ✓ Bits stored as charge in capacitors
- ✓ Absence and presence of charges is interpreted as a binary 0 & 1
- ✓ Charges leak
- ✓ Need refreshing even when powered
- ✓ Simpler construction
- ✓ Smaller per bit
- ✓ Less expensive
- ✓ Need refresh circuits
- ✓ Slower
 - ✓ Most common, cheap
 - ✓ Volatile: must be refreshed (recharged with power) 1000's of times each second
- **Eg: Main memory**
- ✓ Level of charge determines value

Static RAM

- ✓ Bits stored as on/off switches
- ✓ Binary value stores using flip flop
- ✓ Digital
 - ✓ Uses flip-flops
- ✓ No charges to leak
- ✓ No refreshing needed when powered
- ✓ More complex construction
- ✓ More expensive than DRAM
- ✓ Does not need refresh circuits
- ✓ Faster than DRAM
- ✓ **Eg: Cache**

SRAM v DRAM

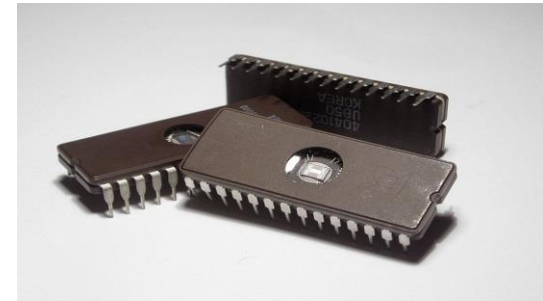
- **Both volatile**
 - Power needed to preserve data
- **Dynamic cell**
 - Simpler to build, smaller
 - More dense
 - Less expensive
 - Needs refresh
 - Larger memory units

Static

- ✓ Less Dense
- ✓ Expensive
- ✓ Faster
- ✓ Cache

Read Only Memory (ROM)

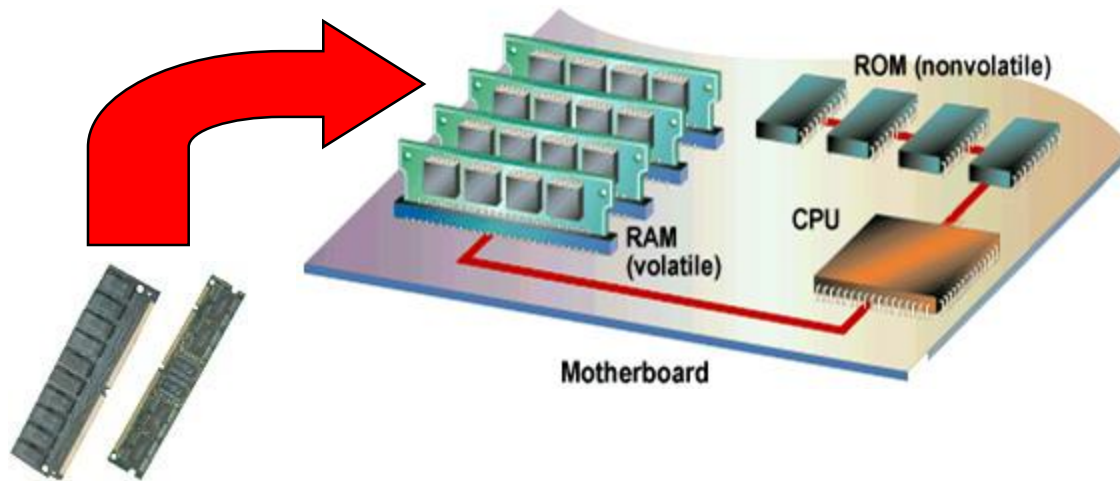
- Permanent storage
 - Nonvolatile
- Data is written during fabrication
- Data insertion step include a relatively large fixed cost whether 1 or thousands copies of a particular ROM are fabricated
- No room for error if one bit is wrong the whole pattern of ROM must be discarded



Applications:-

- Microprogramming
- Library subroutines for frequently wanted functions
- Systems program.
- Function tables

The ROM is nonvolatile because it will not lose its contents when powered down.



RAM plugs into sockets on the motherboard.

Types of ROM

i. Written during manufacture(ROM)

- Very expensive for small runs
- Data and program is permanent available in MM and no need to be loaded from secondary storage devices.

ii. Programmable (once)

- PROM(Can be written only once)
- Needs special equipment to program

iii. Read “mostly”

Useful for application in which Read Operations are far most frequent than write operation.

a. Erasable Programmable (EPROM)

- ✓ Erased by UV light
- ✓ Read and Written electrically
- ✓ Before write operation contents must be erased
- ✓ Nonvolatile
- ✓ EPROM is most expensive than PROM but it has advantage of multiple update memory

b. Electrically Erasable (EEPROM)

- Takes much longer to write than read
- Electrically erasable
- Time to perform write operation is greater than time to perform read operation
- More expensive than EPROM
- Read /write operation is also electrical
- Non volatile
- Eraser is at byte level
- Less denser as compared to EPROM
- It has flexibility of being updateable in place using ordinary bus control address & data lines.

C. Flash memory

- Intermediate between EPROM & EEPROM in terms of both cost & functionality
- Denser than EPROM

Erase whole memory electrically

Refreshing

- Refresh circuit included on chip
- Disable chip
- Count through rows
- Read & Write back
- Takes time
- Slows down apparent performance

Operation of Memory

- Each memory location has a **unique** address
- Address from an instruction is copied to the **MAR** which finds the location in memory
- CPU determines if it is a **store or retrieval**
- Transfer takes place between the **MDR** and memory

Types of Error (Error Correction)

- **Hard Failure**
 - Permanent defect
- **Soft Error**
 - Random, non-destructive
 - No permanent damage to memory
- Detected using Hamming error correcting code

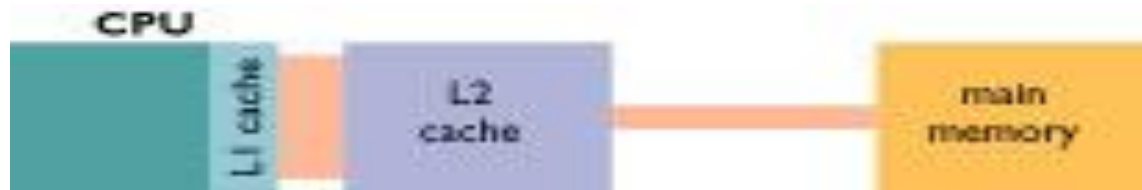
Cache Memory

- Small block of **very fast** temporary memory
- Speeds up **data transfer**
- Stores/keeps **instructions** and **data** used most **frequently** or most **recently**

Types of Cache Memory

- **Internal cache**

- Level 1 (L1)
- Built into microprocessor
- Up to 128KB



- **External cache**

- Level 2 (L2)
- Separate chips
- 256KB or 512 KB
- SRAM technology
- Cheaper and slower than L1
- Faster and more expensive than memory

Advanced DRAM Organization

- Basic DRAM same since first RAM chips
- Enhanced DRAM
 - Contains small SRAM as well
- Cache DRAM
 - Larger SRAM component
 - Use as cache or serial buffer

Synchronous DRAM (SDRAM)

- Access is synchronized with an external clock
- Address is presented to RAM
- RAM finds data (CPU waits in conventional DRAM)
- Since SDRAM moves data in time with system clock, CPU knows when data will be ready
- CPU does not have to wait, it can do something else
- Burst mode allows SDRAM to set up stream of data and fire it out in block
- DDR-SDRAM sends data twice per clock cycle (leading & trailing edge)

DDR SDRAM(Double data rate SDRAM)

- Enhanced version of SDRAM
- SDRAM can only send data once per clock
- Double-data-rate SDRAM can send data twice per clock cycle
 - Once on the Rising edge and once on the falling edge

Characteristics

1. Location
2. Capacity
3. Unit of transfer
4. Access method
5. Performance
6. Physical type
7. Physical characteristics
8. Organisation

Location

- Processor
- (CPU Register, Central Register)
- Internal
- (Primary memory, RAM/ROM, Cache)
- External

Capacity

- Internal memory expressed in (Byte ,KB,MB)
 - External Memory expressed in bytes (Mb,GB,TeraB)
- Word size
 - The natural unit of organisation
- Number of words
 - or Bytes
- Addressable unit
 - Smallest location which can be uniquely addressed
 - Word internally
 - Eg: For N addressable unit
 - $N=2^a$ i.e at least 'a' bit required to uniquely address each location

Unit of Transfer

- **For Main memory** its no of bit read out or written in to memory at a time
- **For External Memory** data is transferred in the form of blocks
- Internal
 - Usually governed by data bus width
- External
 - Usually a block which is much larger than a word

Access Methods (1)

- **Sequential**
 - Start at the beginning and read through in order
 - Access is made in a specific linear sequence.
 - Access time depends on location of data and previous location
 - Access time is highly variable
 - e.g. tape

- **Direct**

- Individual blocks have unique address
- Access is by jumping to vicinity plus sequential search
- Access time depends on location and previous location
- Access time is variable
- e.g. disk

Access Methods (2)

- **Random**

- Individual addresses identify locations exactly at the same amount of time.
- Access time is independent of location or previous access
- e.g. RAM

- **Associative** access memory are content addressable

- Data is located by a comparison with contents of a portion of the store
- Access time is independent of location or previous access
- Access time is constant
- e.g. cache

Performance

- **Access time**

- Time between presenting the address and getting the valid data
- t_0 = Time at which the address appears in address register.
- t_1 = Time at which the desired information appears in the buffer or memory location
- Access time = $t_1 - t_0$

- **Memory Cycle time** is the access time + some additional time before second access can commence

- Time may be required for the memory to “recover” before next access
- Cycle time is access + recovery

Performance

- **Transfer Rate (R)**

For Ram:

- Rate at which data transferred in to or out of the memory unit.
- $R = 1 / \text{Memory cycle time}$

For Non RAM:

$$T_N = T_A + N/R$$

T_N = Average time to read/write N bits

T_A = Average access time

N = No. of bits

R = Transfer rate

Physical Types

- **Semiconductor**
 - RAM/ROM/Cache
- **Magnetic**
 - Disk & Tape
- **Optical**
 - CD & DVD
- **Others**
 - Bubble
 - Hologram

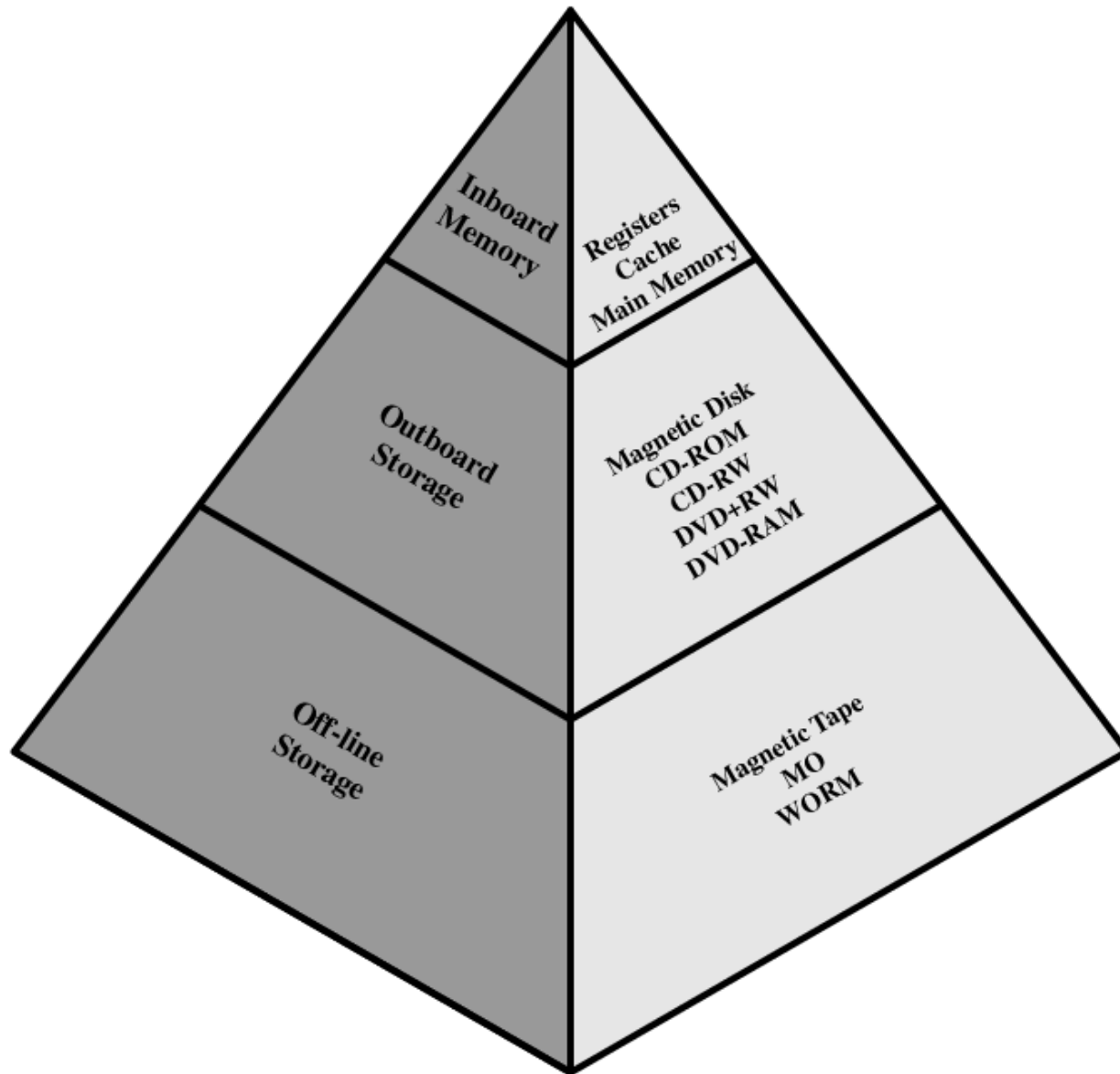
Physical Characteristics

- Decay
- Volatility
- Erasable
- Power consumption

Organisation

- Physical arrangement of bits into words
- Not always obvious
- e.g. interleaved

Memory Hierarchy



Memory Hierarchy

- It is therefore impossible to design an entire system using one technology that offers low cost , high operating speed hence a memory system is designed using different devices to achieve optimum performance
- **Registers**
 - In CPU
- **Internal or Main memory**
 - May include one or more levels of cache
 - “RAM”
- **External memory**
 - Backing store
 - **Faster access time, greater cost/bit**
 - **Greater capacity, small cost/bit**
 - **Greater capacity, slower access time**

Hierarchy List

- Registers
- L1 Cache
- L2 Cache
- Main memory
- Disk cache
- Disk
- Optical
- Tape

Locality of Reference

- Programs tends to access small portion of logical address space at any point of time.

Dimensions

Time , Space , Order

- **Temporal Locality(Locality over Time)**
- **Spatial Locality(Locality Over Space)**
- **Sequential Locality (Locality over Order)**

-
- **Temporal Locality(Locality over Time)**
 - Recently referenced items are likely to be referenced again in ear future eg. Loops ,sub routine
 - **Spatial Locality(Locality Over Space)**
 - Program tend to access item whose addresses are near one another eg. arrays
 - **Sequential Locality (Locality over Order):**Programs instruction are executed sequentially unless specified otherwise.

-
- **Hit Ratio** = Possibility of finding the requested information
 - **Hit ratio=no. of success/no. access**
 - $h_0=0$ $h_n=1$
 - **Miss Ratio** :Probability of not finding the requested information
 - **Miss Ratio=1- h_i**

Memory Optimization

How to minimize effective access time subject to
constrate that $C_T \leq C_O$

Size =Capacity

C_T = Total Cost of Memory Hierarchy

C_O =Upper bound on cost

(The **upper bound** is the smallest value that would round up to the next estimated value)

$C_T = \text{Summation } C_i S_i$

C_i = Cost /byte or words of M_i

S_i = Capacity of memory M_i In bytes or words

Average cost /bit=?